

Number Series

100 Practice Questions

Intermediate to Advanced | Company-style aptitude

How to use this booklet

- Find the missing number (?) in each series and write it in the answer column.
- Work through the parts in order, A to D. Aim for about 30 to 40 seconds per question.
- Use the checking order: differences, then $\times/\div$, then $\times/\div$ with $+/-$, then squares/cubes, then alternate terms.
- Do not look at the solutions until you have finished a whole part.
- Mark your answers against the **Answer Key on the last page**; the worked solutions come just before it.

Part	Questions	Level
A	1 - 25	Intermediate
B	26 - 50	Intermediate to advanced
C	51 - 75	Advanced
D	76 - 100	Advanced, company-level

A note on six questions

Questions 65, 72, 84, 86, 87 and 98 have no clean, unique pattern (they look like misprints or over-engineered series). They are marked * in the solutions. Skip them when you practise, and score yourself out of **94**.

Part A — Intermediate (Q1 to Q25)

Find the missing number (?) in each series.

Q	Series	Your answer
1	7, 12, 19, 28, 39, (?)	
2	4, 9, 19, 39, 79, (?)	
3	3, 8, 15, 24, 35, (?)	
4	11, 17, 26, 38, 53, (?)	
5	6, 13, 27, 55, 111, (?)	
6	2, 6, 12, 20, 30, (?)	
7	5, 11, 23, 47, 95, (?)	
8	10, 18, 30, 46, 66, (?)	
9	1, 4, 10, 22, 46, (?)	
10	8, 15, 24, 35, 48, (?)	
11	13, 21, 34, 55, 89, (?)	
12	2, 5, 11, 23, 47, (?)	
13	9, 16, 25, 36, 49, (?)	
14	3, 12, 27, 48, 75, (?)	
15	4, 10, 22, 46, 94, (?)	
16	7, 14, 28, 56, 112, (?)	
17	15, 22, 31, 42, 55, (?)	
18	1, 3, 7, 15, 31, (?)	
19	20, 17, 13, 8, 2, (?)	
20	2, 9, 28, 65, 126, (?)	
21	6, 18, 54, 162, (?)	
22	81, 27, 9, 3, (?)	
23	5, 8, 14, 26, 50, (?)	
24	12, 23, 45, 89, 177, (?)	
25	1, 8, 27, 64, 125, (?)	

Part B — Intermediate to Advanced (Q26 to Q50)

Find the missing number (?) in each series.

Q	Series	Your answer
26	3, 7, 15, 31, 63, (?)	
27	2, 7, 17, 37, 77, (?)	
28	10, 13, 21, 36, 60, (?)	
29	4, 7, 13, 25, 49, (?)	
30	6, 11, 21, 41, 81, (?)	
31	3, 10, 21, 36, 55, (?)	
32	8, 14, 26, 50, 98, (?)	
33	2, 4, 12, 48, 240, (?)	
34	5, 15, 45, 135, (?)	
35	144, 72, 36, 18, 9, (?)	
36	1, 5, 14, 30, 55, (?)	
37	4, 11, 25, 53, 109, (?)	
38	7, 16, 34, 70, 142, (?)	
39	2, 10, 30, 68, 130, (?)	
40	3, 9, 27, 81, 243, (?)	
41	14, 29, 59, 119, 239, (?)	
42	6, 8, 13, 23, 42, (?)	
43	20, 18, 15, 11, 6, (?)	
44	2, 8, 26, 80, 242, (?)	
45	1, 6, 18, 40, 75, (?)	
46	9, 19, 39, 79, 159, (?)	
47	3, 6, 18, 72, 360, (?)	
48	5, 12, 27, 58, 121, (?)	
49	10, 22, 46, 94, 190, (?)	
50	2, 5, 12, 27, 58, (?)	

Part C — Advanced (Q51 to Q75)

Find the missing number (?) in each series.

Q	Series	Your answer
51	1, 4, 13, 40, 121, (?)	
52	2, 6, 18, 54, 162, (?)	
53	3, 8, 18, 38, 78, (?)	
54	5, 14, 32, 68, 140, (?)	
55	7, 19, 43, 91, 187, (?)	
56	2, 9, 28, 65, 126, (?)	
57	4, 16, 64, 256, (?)	
58	1, 7, 25, 79, 241, (?)	
59	6, 17, 40, 87, 182, (?)	
60	3, 11, 35, 107, 323, (?)	
61	8, 17, 35, 71, 143, (?)	
62	2, 12, 36, 80, 150, (?)	
63	5, 9, 19, 39, 79, (?)	
64	10, 20, 44, 92, 188, (?)	
65	3, 13, 39, 93, 189, (?)	
66	1, 2, 6, 24, 120, (?)	
67	2, 6, 24, 120, 720, (?)	
68	9, 18, 36, 72, 144, (?)	
69	7, 15, 31, 63, 127, (?)	
70	4, 13, 40, 121, 364, (?)	
71	11, 23, 47, 95, 191, (?)	
72	2, 11, 38, 101, 226, (?)	
73	6, 15, 32, 57, 90, (?)	
74	1, 9, 36, 100, 225, (?)	
75	8, 27, 64, 125, 216, (?)	

Part D — Advanced, Company-Level (Q76 to Q100)

Find the missing number (?) in each series.

Q	Series	Your answer
76	2, 5, 14, 41, 122, (?)	
77	3, 10, 31, 94, 283, (?)	
78	4, 9, 19, 39, 79, (?)	
79	6, 19, 58, 175, 526, (?)	
80	5, 16, 49, 148, 445, (?)	
81	1, 5, 17, 53, 161, (?)	
82	2, 7, 22, 67, 202, (?)	
83	9, 28, 85, 256, 769, (?)	
84	3, 12, 48, 192, 960, (?)	
85	4, 20, 120, 840, 6720, (?)	
86	2, 8, 32, 160, 960, (?)	
87	7, 12, 24, 51, 103, (?)	
88	5, 13, 29, 61, 125, (?)	
89	10, 21, 43, 87, 175, (?)	
90	1, 3, 8, 18, 38, (?)	
91	2, 6, 15, 31, 56, (?)	
92	4, 10, 24, 52, 108, (?)	
93	6, 14, 30, 62, 126, (?)	
94	3, 7, 16, 34, 69, (?)	
95	2, 4, 10, 28, 82, (?)	
96	5, 11, 24, 50, 101, (?)	
97	8, 19, 42, 89, 184, (?)	
98	1, 6, 20, 55, 126, (?)	
99	7, 22, 67, 202, 607, (?)	
100	2, 9, 28, 65, 126, 217, (?)	

Worked Solutions

Pattern and working for every question. Questions marked * have no clean unique pattern.

Part A — Intermediate

Q	Series	Answer	Pattern and working
1	7, 12, 19, 28, 39, ?	52	Differences +5, +7, +9, +11 (rise by 2) Next +13: $39 + 13 = 52$
2	4, 9, 19, 39, 79, ?	159	$\times 2 + 1$ $79 \times 2 + 1 = 159$
3	3, 8, 15, 24, 35, ?	48	$n^2 - 1$ (2^2-1 , 3^2-1 , ...) $7^2 - 1 = 48$
4	11, 17, 26, 38, 53, ?	71	Differences +6, +9, +12, +15 (rise by 3) Next +18: $53 + 18 = 71$
5	6, 13, 27, 55, 111, ?	223	$\times 2 + 1$ $111 \times 2 + 1 = 223$
6	2, 6, 12, 20, 30, ?	42	$n(n+1)$: 1×2 , 2×3 , 3×4 , 4×5 , 5×6 $6 \times 7 = 42$
7	5, 11, 23, 47, 95, ?	191	$\times 2 + 1$ $95 \times 2 + 1 = 191$
8	10, 18, 30, 46, 66, ?	90	Differences +8, +12, +16, +20 (rise by 4) Next +24: $66 + 24 = 90$
9	1, 4, 10, 22, 46, ?	94	$\times 2 + 2$ $46 \times 2 + 2 = 94$
10	8, 15, 24, 35, 48, ?	63	Differences +7, +9, +11, +13 Next +15: $48 + 15 = 63$
11	13, 21, 34, 55, 89, ?	144	Fibonacci type: sum of the previous two $55 + 89 = 144$
12	2, 5, 11, 23, 47, ?	95	$\times 2 + 1$ $47 \times 2 + 1 = 95$
13	9, 16, 25, 36, 49, ?	64	Consecutive squares 3^2 to 7^2 $8^2 = 64$
14	3, 12, 27, 48, 75, ?	108	$3 \times n^2$: 3×1 , 3×4 , 3×9 , 3×16 , 3×25 $3 \times 36 = 108$
15	4, 10, 22, 46, 94, ?	190	$\times 2 + 2$ $94 \times 2 + 2 = 190$
16	7, 14, 28, 56, 112, ?	224	$\times 2$ $112 \times 2 = 224$
17	15, 22, 31, 42, 55, ?	70	Differences +7, +9, +11, +13 Next +15: $55 + 15 = 70$
18	1, 3, 7, 15, 31, ?	63	$\times 2 + 1$ $31 \times 2 + 1 = 63$
19	20, 17, 13, 8, 2, ?	-5	Differences -3, -4, -5, -6 Next -7: $2 - 7 = -5$
20	2, 9, 28, 65, 126, ?	217	$n^3 + 1$ $6^3 + 1 = 217$

Q	Series	Answer	Pattern and working
21	6, 18, 54, 162, ?	486	$\times 3$ $162 \times 3 = 486$
22	81, 27, 9, 3, ?	1	$\div 3$ $3 \div 3 = 1$
23	5, 8, 14, 26, 50, ?	98	Differences +3, +6, +12, +24 (double) Next +48: $50 + 48 = 98$
24	12, 23, 45, 89, 177, ?	353	$\times 2 - 1$ $177 \times 2 - 1 = 353$
25	1, 8, 27, 64, 125, ?	216	Cubes 1^3 to 5^3 $6^3 = 216$

Part B — Intermediate to Advanced

Q	Series	Answer	Pattern and working
26	3, 7, 15, 31, 63, ?	127	$\times 2 + 1$ $63 \times 2 + 1 = 127$
27	2, 7, 17, 37, 77, ?	157	$\times 2 + 3$ $77 \times 2 + 3 = 157$
28	10, 13, 21, 36, 60, ?	95	Differences 3, 8, 15, 24 = 2^2-1 , 3^2-1 , 4^2-1 , 5^2-1 Next $6^2 - 1 = 35$: $60 + 35 = 95$
29	4, 7, 13, 25, 49, ?	97	$\times 2 - 1$ $49 \times 2 - 1 = 97$
30	6, 11, 21, 41, 81, ?	161	$\times 2 - 1$ $81 \times 2 - 1 = 161$
31	3, 10, 21, 36, 55, ?	78	Differences +7, +11, +15, +19 (rise by 4) Next +23: $55 + 23 = 78$
32	8, 14, 26, 50, 98, ?	194	$\times 2 - 2$ $98 \times 2 - 2 = 194$
33	2, 4, 12, 48, 240, ?	1440	Multipliers $\times 2$, $\times 3$, $\times 4$, $\times 5$ Next $\times 6$: $240 \times 6 = 1440$
34	5, 15, 45, 135, ?	405	$\times 3$ $135 \times 3 = 405$
35	144, 72, 36, 18, 9, ?	4.5	$\div 2$ $9 \div 2 = 4.5$
36	1, 5, 14, 30, 55, ?	91	Differences are squares: +4, +9, +16, +25 Next $6^2 = 36$: $55 + 36 = 91$
37	4, 11, 25, 53, 109, ?	221	$\times 2 + 3$ $109 \times 2 + 3 = 221$
38	7, 16, 34, 70, 142, ?	286	$\times 2 + 2$ $142 \times 2 + 2 = 286$
39	2, 10, 30, 68, 130, ?	222	$n^3 + n$ $6^3 + 6 = 222$
40	3, 9, 27, 81, 243, ?	729	$\times 3$ $243 \times 3 = 729$
41	14, 29, 59, 119, 239, ?	479	$\times 2 + 1$ $239 \times 2 + 1 = 479$

Q	Series	Answer	Pattern and working
42	6, 8, 13, 23, 42, ?	78	Differences 2, 5, 10, 19; these rise by 3, 5, 9 (each $2^n + 1$) Next rise 17, so next difference 36: $42 + 36 = 78$
43	20, 18, 15, 11, 6, ?	0	Differences -2, -3, -4, -5 Next -6: $6 - 6 = 0$
44	2, 8, 26, 80, 242, ?	728	$\times 3 + 2$ $242 \times 3 + 2 = 728$
45	1, 6, 18, 40, 75, ?	126	$n \times \text{triangular number}$: $1 \times 1, 2 \times 3, 3 \times 6, 4 \times 10, 5 \times 15$ $6 \times 21 = 126$
46	9, 19, 39, 79, 159, ?	319	$\times 2 + 1$ $159 \times 2 + 1 = 319$
47	3, 6, 18, 72, 360, ?	2160	Multipliers $\times 2, \times 3, \times 4, \times 5$ Next $\times 6$: $360 \times 6 = 2160$
48	5, 12, 27, 58, 121, ?	248	$\times 2 + 2, \times 2 + 3, \times 2 + 4, \times 2 + 5$ Next $\times 2 + 6$: $121 \times 2 + 6 = 248$
49	10, 22, 46, 94, 190, ?	382	$\times 2 + 2$ $190 \times 2 + 2 = 382$
50	2, 5, 12, 27, 58, ?	121	$\times 2 + 1, \times 2 + 2, \times 2 + 3, \times 2 + 4$ Next $\times 2 + 5$: $58 \times 2 + 5 = 121$

Part C — Advanced

Q	Series	Answer	Pattern and working
51	1, 4, 13, 40, 121, ?	364	$\times 3 + 1$ $121 \times 3 + 1 = 364$
52	2, 6, 18, 54, 162, ?	486	$\times 3$ $162 \times 3 = 486$
53	3, 8, 18, 38, 78, ?	158	$\times 2 + 2$ $78 \times 2 + 2 = 158$
54	5, 14, 32, 68, 140, ?	284	$\times 2 + 4$ $140 \times 2 + 4 = 284$
55	7, 19, 43, 91, 187, ?	379	$\times 2 + 5$ $187 \times 2 + 5 = 379$
56	2, 9, 28, 65, 126, ?	217	$n^3 + 1$ $6^3 + 1 = 217$
57	4, 16, 64, 256, ?	1024	$\times 4$ $256 \times 4 = 1024$
58	1, 7, 25, 79, 241, ?	727	$\times 3 + 4$ $241 \times 3 + 4 = 727$
59	6, 17, 40, 87, 182, ?	373	$\times 2 + 5, \times 2 + 6, \times 2 + 7, \times 2 + 8$ Next $\times 2 + 9$: $182 \times 2 + 9 = 373$
60	3, 11, 35, 107, 323, ?	971	$\times 3 + 2$ $323 \times 3 + 2 = 971$
61	8, 17, 35, 71, 143, ?	287	$\times 2 + 1$ $143 \times 2 + 1 = 287$

Q	Series	Answer	Pattern and working
62	2, 12, 36, 80, 150, ?	252	$n^2 \times (n+1)$: $1 \times 2, 4 \times 3, 9 \times 4, 16 \times 5, 25 \times 6$ $36 \times 7 = 252$
63	5, 9, 19, 39, 79, ?	159	Differences 4, 10, 20, 40 (double after the first); same as $\times 2 + 1$ from 9 onward Next +80: $79 + 80 = 159$
64	10, 20, 44, 92, 188, ?	380	Differences 10, 24, 48, 96 (double after the first); same as $\times 2 + 4$ from 20 onward Next +192: $188 + 192 = 380$
65*	3, 13, 39, 93, 189, ?	n/a*	No clean rule Differences 10, 26, 54, 96 give 16, 28, 42, then 12, 14: no exam-style pattern. Skip it.
66	1, 2, 6, 24, 120, ?	720	Factorials $1!$ to $5!$ $6! = 720$
67	2, 6, 24, 120, 720, ?	5040	Factorials $2!$ to $6!$ $7! = 5040$
68	9, 18, 36, 72, 144, ?	288	$\times 2$ $144 \times 2 = 288$
69	7, 15, 31, 63, 127, ?	255	$\times 2 + 1$ $127 \times 2 + 1 = 255$
70	4, 13, 40, 121, 364, ?	1093	$\times 3 + 1$ $364 \times 3 + 1 = 1093$
71	11, 23, 47, 95, 191, ?	383	$\times 2 + 1$ $191 \times 2 + 1 = 383$
72*	2, 11, 38, 101, 226, ?	n/a*	No clean rule Differences 9, 27, 63, 125 give 18, 36, 62, then 18, 26: no exam-style pattern. Skip it.
73	6, 15, 32, 57, 90, ?	131	Differences +9, +17, +25, +33 (rise by 8) Next +41: $90 + 41 = 131$
74	1, 9, 36, 100, 225, ?	441	Squares of triangular numbers: $1^2, 3^2, 6^2, 10^2, 15^2$ Next root 21: $21^2 = 441$
75	8, 27, 64, 125, 216, ?	343	Cubes 2^3 to 6^3 $7^3 = 343$

Part D — Advanced, Company-Level

Q	Series	Answer	Pattern and working
76	2, 5, 14, 41, 122, ?	365	$\times 3 - 1$ $122 \times 3 - 1 = 365$
77	3, 10, 31, 94, 283, ?	850	$\times 3 + 1$ $283 \times 3 + 1 = 850$
78	4, 9, 19, 39, 79, ?	159	$\times 2 + 1$ $79 \times 2 + 1 = 159$
79	6, 19, 58, 175, 526, ?	1579	$\times 3 + 1$ $526 \times 3 + 1 = 1579$
80	5, 16, 49, 148, 445, ?	1336	$\times 3 + 1$ $445 \times 3 + 1 = 1336$

Q	Series	Answer	Pattern and working
81	1, 5, 17, 53, 161, ?	485	$\times 3 + 2$ $161 \times 3 + 2 = 485$
82	2, 7, 22, 67, 202, ?	607	$\times 3 + 1$ $202 \times 3 + 1 = 607$
83	9, 28, 85, 256, 769, ?	2308	$\times 3 + 1$ $769 \times 3 + 1 = 2308$
84*	3, 12, 48, 192, 960, ?	n/a*	No clean rule (likely a misprint) Multipliers are $\times 4$, $\times 4$, $\times 4$, $\times 5$ with no reason for the jump. If the last term were 768 ($\times 4$), the answer would be 3072.
85	4, 20, 120, 840, 6720, ?	60480	Multipliers $\times 5$, $\times 6$, $\times 7$, $\times 8$ Next $\times 9$: $6720 \times 9 = 60480$
86*	2, 8, 32, 160, 960, ?	n/a*	No clean rule Multipliers $\times 4$, $\times 4$, $\times 5$, $\times 6$: the repeated 4 breaks the run, so there is no unique answer. Skip it.
87*	7, 12, 24, 51, 103, ?	n/a*	No clean rule Differences 5, 12, 27, 52 give 7, 15, 25, then 8, 10: no exam-style pattern. Skip it.
88	5, 13, 29, 61, 125, ?	253	$\times 2 + 3$ $125 \times 2 + 3 = 253$
89	10, 21, 43, 87, 175, ?	351	$\times 2 + 1$ $175 \times 2 + 1 = 351$
90	1, 3, 8, 18, 38, ?	78	Differences 2, 5, 10, 20 (double after the first) Next $+40$: $38 + 40 = 78$
91	2, 6, 15, 31, 56, ?	92	Differences are squares: $+2^2$, $+3^2$, $+4^2$, $+5^2$ Next $6^2 = 36$: $56 + 36 = 92$
92	4, 10, 24, 52, 108, ?	220	Differences 6, 14, 28, 56 (double after the first) Next $+112$: $108 + 112 = 220$
93	6, 14, 30, 62, 126, ?	254	$\times 2 + 2$ $126 \times 2 + 2 = 254$
94	3, 7, 16, 34, 69, ?	137	Differences 4, 9, 18, 35; these rise by 5, 9, 17 (each $2^n + 1$) Next rise 33, so next difference 68: $69 + 68 = 137$
95	2, 4, 10, 28, 82, ?	244	$\times 3 - 2$ $82 \times 3 - 2 = 244$
96	5, 11, 24, 50, 101, ?	201	Differences 6, 13, 26, 51; these rise by 7, 13, 25 (each $= 2 \times \text{previous} - 1$) Next rise 49, so next difference 100: $101 + 100 = 201$
97	8, 19, 42, 89, 184, ?	375	$\times 2 + 3$, $\times 2 + 4$, $\times 2 + 5$, $\times 2 + 6$ Next $\times 2 + 7$: $184 \times 2 + 7 = 375$
98*	1, 6, 20, 55, 126, ?	n/a*	No clean rule Differences 5, 14, 35, 71 give 9, 21, 36, then 12, 15: no exam-style pattern. Skip it.
99	7, 22, 67, 202, 607, ?	1822	$\times 3 + 1$ $607 \times 3 + 1 = 1822$
100	2, 9, 28, 65, 126, 217, ?	344	$n^3 + 1$ $7^3 + 1 = 344$

Answer Key

All 100 answers at a glance. * = no clean unique pattern (skip; score out of 94).

1. 52	26. 127	51. 364	76. 365
2. 159	27. 157	52. 486	77. 850
3. 48	28. 95	53. 158	78. 159
4. 71	29. 97	54. 284	79. 1579
5. 223	30. 161	55. 379	80. 1336
6. 42	31. 78	56. 217	81. 485
7. 191	32. 194	57. 1024	82. 607
8. 90	33. 1440	58. 727	83. 2308
9. 94	34. 405	59. 373	84. n/a*
10. 63	35. 4.5	60. 971	85. 60480
11. 144	36. 91	61. 287	86. n/a*
12. 95	37. 221	62. 252	87. n/a*
13. 64	38. 286	63. 159	88. 253
14. 108	39. 222	64. 380	89. 351
15. 190	40. 729	65. n/a*	90. 78
16. 224	41. 479	66. 720	91. 92
17. 70	42. 78	67. 5040	92. 220
18. 63	43. 0	68. 288	93. 254
19. -5	44. 728	69. 255	94. 137
20. 217	45. 126	70. 1093	95. 244
21. 486	46. 319	71. 383	96. 201
22. 1	47. 2160	72. n/a*	97. 375
23. 98	48. 248	73. 131	98. n/a*
24. 353	49. 382	74. 441	99. 1822
25. 216	50. 121	75. 343	100. 344